

Paley two-edge switching for proper transposed sesqui arrays

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Abstract

Let $q \equiv 3 \pmod{4}$ be an odd prime power. We construct, for every $q \geq 11$, a proper transposed sesqui array

$$SA^T\left(2q, \frac{q+1}{2}, -, \frac{q-1}{2}, \frac{q+1}{2} : q \times (q+1)\right).$$

The construction starts from a Paley matching between the nonzero squares and nonsquares of $\mathbb{F}_q$, then crosses two matching edges. Five quadratic character conditions preserve the column–column and row–column concurrences while a nonzero second-moment defect forces the row–row concurrence to be nonconstant. An exact character count reduces the existence of suitable switching parameters to the Hasse bound for a family of elliptic curves. The bound covers $q \geq 19$, and the remaining case $q = 11$ has an explicit witness.

Status. This is a proof-complete initial draft. Its symbolic identities, finite-field instances, and boundary cases have passed implementation-separated internal checks. External mathematical review and a definitive priority assessment remain open, so the novelty statement below is qualified.

1 Introduction

Row–column designs are binary arrays whose rows and columns are regarded as blocks on a common symbol set. Triple arrays require constant row–row, column–column, and row–column concurrence. A transposed sesqui array keeps the latter two conditions but does not require constant row–row concurrence; it is *proper* when row–row concurrence is in fact nonconstant [2].

For odd prime powers $q \equiv 3 \pmod{4}$ with $q > 3$, the classical Paley construction gives triple arrays with parameters

$$TA\left(2q, \frac{q+1}{2}, \frac{q+1}{2}, \frac{q-1}{2}, \frac{q+1}{2} : q \times (q+1)\right)$$

[7, 5]. Thus the dimensions, replication, and concurrence parameters studied here are not new. Our objective is to retain the Paley column structure while replacing the constant row concurrence by a provably nonconstant one.

The construction is local. A standard Paley matching pairs every nonzero square with a non-square. We cross two of its edges, subject to five quadratic character conditions. This changes only four endpoints, but it has three global consequences: the sum map remains two-to-one away from zero, the column design remains Paley, and the row block acquires a nonzero moment defect. The first two properties give a transposed sesqui array; the third proves that it is proper.

The nonlocal part of the argument is existence. For a fixed matching parameter α , the number of valid second edges is expressed exactly as a cubic quadratic-character sum. The associated curve is elliptic, and the Hasse bound makes the count positive for every $q \geq 19$. A direct witness closes $q = 11$. No finite computation is used to infer the infinite family.

The smallest instance has $(v, r, c) = (22, 11, 12)$. The 2025 enumeration and construction table of Jäger et al. marks its proper transposed-sesqui entry as $?, T$: a same-parameter triple array was known, while their search and listed constructions did not produce a proper example [2, Table 2]. To the best of our knowledge, the Paley two-edge switching construction and the resulting proper family have not appeared in the row-column design literature.

2 Definitions and Paley preliminaries

Let A be an $r \times c$ array on a symbol set Ω of size v . Write R_i and C_j for the symbol supports of row i and column j .

Definition 2.1. The array A is *binary* if no symbol occurs twice in one row or one column, and *equireplicate* with replication e if every symbol occurs in exactly e cells. It is a transposed sesqui array if there are constants λ_{CC} and λ_{RC} such that

$$|C_j \cap C_{j'}| = \lambda_{CC} \quad (j \neq j'), \quad |R_i \cap C_j| = \lambda_{RC} \quad \text{for all } i, j.$$

We write

$$SA^\Gamma(v, e, -, \lambda_{CC}, \lambda_{RC} : r \times c).$$

It is *proper* if $|R_i \cap R_{i'}|$ is not constant over distinct rows.

Let $F = \mathbb{F}_q$, where throughout $q \equiv 3 \pmod{4}$ is an odd prime power. Let $\chi : F \rightarrow \{0, \pm 1\}$ be the quadratic character, with $\chi(0) = 0$, and put

$$Q = \{x^2 : x \in F^*\}, \quad N = F^* \setminus Q, \quad N_0 = N \cup \{0\}.$$

Then $\chi(-1) = -1$, and $Q \sqcup N_0 = F$.

Lemma 2.2 (Paley intersections). *For every $h \in F^*$,*

$$|Q \cap (Q + h)| = \frac{q-3}{4}, \quad |N_0 \cap (N_0 + h)| = \frac{q+1}{4}.$$

In particular, Q is a $(q, (q-1)/2, (q-3)/4)$ difference set in the additive group of F , and N_0 is its complement.

Proof. For $h \neq 0$, the standard quadratic sum

$$\sum_{x \in F} \chi(x(x-h)) = -1$$

follows by completing the square. Indeed, after scaling by h , it reduces to $\sum_t \chi(t^2 - 1) = -1$; the equation $y^2 = t^2 - 1$ has $q-1$ solutions because $(t-y)(t+y) = 1$, and counting its solutions by t gives the stated sum. Hence

$$\frac{1}{4} \sum_{x \in F} (1 + \chi(x))(1 + \chi(x-h)) = \frac{q-1}{4}.$$

This product is the indicator of $Q \cap (Q + h)$, except at $x = 0, h$. The two false contributions total

$$\frac{1 + \chi(-h)}{4} + \frac{1 + \chi(h)}{4} = \frac{1}{2},$$

because $\chi(-1) = -1$. Thus $|Q \cap (Q + h)| = (q - 3)/4$. Since $N_0 = F \setminus Q$, taking complements gives $|N_0 \cap (N_0 + h)| = (q + 1)/4$. $\square$

3 A switched Paley matching

We first isolate the conditions on the two crossed edges.

Proposition 3.1 (Two-edge switch). *Suppose $\alpha, r \in F$ satisfy*

$$\chi(\alpha) = \chi(1 + \alpha) = -1, \tag{1}$$

and

$$\chi(r) = \chi(1 + \alpha r) = \chi(r + \alpha) = 1. \tag{2}$$

Define $\pi_0 : F \rightarrow F$ by

$$\pi_0(0) = 0, \quad \pi_0(u) = \alpha u \ (u \in Q), \quad \pi_0(v) = \alpha^{-1}v \ (v \in N). \tag{3}$$

Replace the two pairs

$$\{1, \alpha\}, \quad \{r, \alpha r\}$$

by

$$\{1, \alpha r\}, \quad \{r, \alpha\},$$

and leave every other pair of π_0 unchanged. The resulting map π is an involution. Moreover, the map

$$s : F \longrightarrow F, \quad s(b) = b + \pi(b), \tag{4}$$

has one singleton fibre $\{0\}$ over zero and $(q - 1)/2$ two-element fibres over distinct nonzero values. Each two-element fibre contains one square and one nonsquare.

Proof. Equation (3) pairs each member of Q with a member of N , and conversely, because $\chi(\alpha) = -1$. The condition $r = 1$ is impossible: it would make $\chi(1 + \alpha r) = \chi(1 + \alpha) = -1$, contrary to (2). Hence $1, r, \alpha, \alpha r$ are four distinct nonzero endpoints, and the crossed pairs still join Q to N .

Every unchanged pair has sum $(1 + \alpha)u$ for some $u \in Q \setminus \{1, r\}$. These sums are distinct and lie in N by (1). The two new sums are

$$1 + \alpha r, \quad r + \alpha,$$

which lie in Q by (2); hence they cannot collide with an unchanged sum. They do not collide with one another because

$$(1 + \alpha r) - (r + \alpha) = (1 - \alpha)(1 - r) \neq 0.$$

Together with the fixed point 0, this proves all assertions about the fibres of s . $\square$

Let

$$D = s(F) = \{b + \pi(b) : b \in F\}. \tag{5}$$

Proposition 3.1 gives $|D| = (q + 1)/2$. Relative to the unswitched sum set N_0 , it has the explicit form

$$D = (N_0 \setminus \{1 + \alpha, r(1 + \alpha)\}) \cup \{1 + \alpha r, r + \alpha\}. \tag{6}$$

4 The cellwise array construction

The next step records the ordering, including the zero fibre and the extra column.

Theorem 4.1 (Array from the switched matching). *Under the hypotheses of Proposition 3.1, there is a binary equireplicate transposed sesqui array*

$$SA^T\left(2q, \frac{q+1}{2}, -, \frac{q-1}{2}, \frac{q+1}{2} : q \times (q+1)\right).$$

Its row-row concurrence is nonconstant if and only if the additive intersection numbers $|D \cap (D+h)|$, $h \in F^$, are nonconstant.*

Proof. Take the symbol set $\Omega = F \times \{0, 1\}$, index rows by $a \in F$, and index columns by $F \cup \{\infty\}$. Define

$$\ell(b) = \begin{cases} 0, & \pi(b) \in Q, \\ 1, & \pi(b) \in N_0, \end{cases}$$

and, for a finite column y , put $b = y - a$ and

$$A(a, y) = (a + b + \pi(b), \ell(b)). \quad (7)$$

In the additional column set

$$A(a, \infty) = (a, 0). \quad (8)$$

Fix a row a . Each nonzero fibre of s contains a square and a nonsquare, which receive different labels because π interchanges the two character classes. Thus every $d \in D \setminus \{0\}$ contributes both $(a + d, 0)$ and $(a + d, 1)$. The zero fibre is $b = 0$: its finite cell is $(a, 1)$, while (8) supplies $(a, 0)$. Therefore the row support is exactly

$$R_a = (a + D) \times \{0, 1\}. \quad (9)$$

In particular, every row is binary.

Fix a finite column y . As a runs through F , the element $b = y - a$ runs through F , and $\pi(b)$ also runs through F . Since the first coordinate in (7) is $y + \pi(b)$, the column support is

$$C_y = ((y + Q) \times \{0\}) \cup ((y + N_0) \times \{1\}), \quad (10)$$

while

$$C_\infty = F \times \{0\}. \quad (11)$$

These columns are binary. A symbol with label zero occurs in $|Q|$ finite columns and the infinity column; a symbol with label one occurs in $|N_0|$ finite columns. Hence every symbol has replication $(q+1)/2$.

For two finite columns, Lemma 2.2 and the disjoint labels give

$$|C_y \cap C_{y'}| = \frac{q-3}{4} + \frac{q+1}{4} = \frac{q-1}{2}.$$

A finite column meets C_∞ in its $|Q| = (q-1)/2$ label-zero symbols, so column concurrence is constant. Finally, $Q \sqcup N_0 = F$ implies

$$|R_a \cap C_y| = |(a + D) \cap (y + Q)| + |(a + D) \cap (y + N_0)| = |D| = \frac{q+1}{2},$$

and the same value follows from (11). This proves the transposed-sesqui conditions.

For $h \neq 0$, equation (9) gives

$$|R_a \cap R_{a+h}| = 2|D \cap (D + h)|. \quad (12)$$

The final assertion follows. $\square$

5 A moment obstruction to constant row concurrence

We now show that the two-edge switch forces the array in Theorem 4.1 to be proper.

Lemma 5.1 (Difference-set moment identity). *Let $E \subseteq F$, where $q > 3$ is odd, and set*

$$k = |E|, \quad S_1(E) = \sum_{x \in E} x, \quad S_2(E) = \sum_{x \in E} x^2.$$

If $|E \cap (E + h)|$ is constant for $h \in F^$, then*

$$kS_2(E) - S_1(E)^2 = 0 \tag{13}$$

in F .

Proof. Constant nonzero translation intersections make E a difference set in the additive group of F ; write its difference multiplicity as λ . Summing the square of every ordered difference gives

$$\sum_{x, y \in E} (x - y)^2 = 2kS_2(E) - 2S_1(E)^2 = \lambda \sum_{h \in F^*} h^2.$$

The last field sum is zero. For example, choose $c \in F^*$ with $c^2 \neq 1$; multiplication by c permutes F^* , so the sum equals c^2 times itself. Since the characteristic is odd, division by two proves (13). $\square$

Proposition 5.2 (Nonzero switched moment). *The block D in (6) satisfies*

$$|D|S_2(D) - S_1(D)^2 = -\alpha(r - 1)^2 \neq 0. \tag{14}$$

Consequently, the array in Theorem 4.1 is proper.

Proof. The removed and inserted elements in (6) have the same sum:

$$(1 + \alpha) + r(1 + \alpha) = (1 + \alpha r) + (r + \alpha).$$

Thus the first moment does not change. Direct expansion gives the change in the second moment:

$$\begin{aligned} \Delta S_2 &= (1 + \alpha r)^2 + (r + \alpha)^2 - (1 + \alpha)^2 - r^2(1 + \alpha)^2 \\ &= -2\alpha(r - 1)^2. \end{aligned}$$

The original block N_0 is a difference set, so its moment defect is zero by Lemma 5.1. In the field F , the integer $|D| = (q + 1)/2$ equals $1/2$, because $q = 0$ in F . Hence the new defect is

$$|D|\Delta S_2 = -\alpha(r - 1)^2,$$

which is nonzero. Lemma 5.1 shows that the translation intersections of D cannot be constant; now use (12). $\square$

6 Existence of switching parameters

It remains to prove that α, r satisfying (1)–(2) always exist in the claimed range.

Lemma 6.1 (The first parameter). *Let*

$$\mathcal{A} = \{\alpha \in F : \chi(\alpha) = \chi(1 + \alpha) = -1\}.$$

Then

$$|\mathcal{A}| = \frac{q-3}{4}. \quad (15)$$

Proof. The standard quadratic-character sum is

$$\sum_{\alpha \in F} \chi(\alpha(1 + \alpha)) = -1.$$

Expanding $(1 - \chi(\alpha))(1 - \chi(1 + \alpha))/4$ and summing therefore gives $(q-1)/4$. This expression is the indicator of membership in $\mathcal{A}$ except at the zeros. At $\alpha = 0$ it contributes zero, while at $\alpha = -1$ it contributes $1/2$, although $-1 \notin \mathcal{A}$. Removing this false contribution proves (15). $\square$

Fix $\alpha \in \mathcal{A}$, and define

$$C(\alpha) = \#\{r \in F : \chi(r) = \chi(1 + \alpha r) = \chi(r + \alpha) = 1\}, \quad (16)$$

$$T(\alpha) = \sum_{r \in F} \chi(r(1 + \alpha r)(r + \alpha)). \quad (17)$$

Lemma 6.2 (Exact character count). *For every $\alpha \in \mathcal{A}$,*

$$C(\alpha) = \frac{q-3+T(\alpha)-4\chi(\alpha-1)}{8}. \quad (18)$$

Proof. Temporarily sum the product of the three positive-character indicators:

$$\frac{1}{8} \sum_{r \in F} (1 + \chi(r))(1 + \chi(1 + \alpha r))(1 + \chi(r + \alpha)).$$

The three linear character sums vanish. For a quadratic polynomial with nonzero leading coefficient and nonzero discriminant, completing the square gives $\sum_x \chi(Ax^2 + Bx + C) = -\chi(A)$. Since $\alpha \notin \{0, \pm 1\}$, this identity gives the three pairwise sums

$$\begin{aligned} \sum_r \chi(r(1 + \alpha r)) &= 1, \\ \sum_r \chi(r(r + \alpha)) &= -1, \\ \sum_r \chi((1 + \alpha r)(r + \alpha)) &= 1. \end{aligned}$$

Together with (17), the uncorrected total is $(q + 1 + T(\alpha))/8$.

The zero $r = 0$ contributes nothing to that expanded expression. The two other zeros, $r = -1/\alpha$ and $r = -\alpha$, are distinct and each contributes falsely

$$\frac{1 + \chi(\alpha - 1)}{4}.$$

Indeed, the remaining nonzero character is $\chi(\alpha - 1)$ at each point, while the other nonzero factor has character one. Subtracting both contributions gives (18). $\square$

Lemma 6.3 (Elliptic-curve bound). *For every $\alpha \in \mathcal{A}$,*

$$|T(\alpha)| \leq 2\sqrt{q}. \quad (19)$$

Consequently,

$$C(\alpha) \geq \frac{q - 7 - 2\sqrt{q}}{8}. \quad (20)$$

Proof. Consider

$$E_\alpha : z^2 = r(1 + \alpha r)(r + \alpha). \quad (21)$$

The roots $0, -1/\alpha, -\alpha$ are pairwise distinct because $\alpha \notin \{0, \pm 1\}$. Hence the affine cubic is nonsingular. Its projective closure

$$Z^2W = R(W + \alpha R)(R + \alpha W)$$

has the unique point $[0 : 1 : 0]$ at infinity, and the partial derivative with respect to W is nonzero there. Thus E_α is an elliptic curve over F .

For each $r \in F$, the number of affine z -values is $1 + \chi(r(1 + \alpha r)(r + \alpha))$. Including the point at infinity gives

$$\#E_\alpha(F) = q + 1 + T(\alpha).$$

The Hasse bound for elliptic curves over finite fields [8, Chapter V] proves (19). Substituting $T(\alpha) \geq -2\sqrt{q}$ and $\chi(\alpha - 1) \leq 1$ into (18) proves (20). $\square$

Proposition 6.4 (Parameter existence). *If $q \equiv 3 \pmod{4}$ is an odd prime power and $q \geq 19$, then there are $\alpha, r \in F$ satisfying (1)–(2).*

Proof. Lemma 6.1 gives $\mathcal{A} \neq \emptyset$. For any $\alpha \in \mathcal{A}$, the right-hand side of (20) is positive when $q \geq 19$. Since $C(\alpha)$ is an integer, a valid r exists. As observed in Proposition 3.1, its conditions automatically exclude $r = 1$. $\square$

7 The boundary instance and the main theorem

For $q = 11$, take $\alpha = 7$ and $r = 5$. The involution consists of the fixed point 0 and the pairs

$$\{1, 2\}, \quad \{5, 7\}, \quad \{3, 10\}, \quad \{4, 6\}, \quad \{9, 8\}.$$

All five character conditions hold. Its sum set is

$$D = \{0, 1, 2, 3, 6, 10\} \subset \mathbb{F}_{11}, \quad (22)$$

and (14) equals

$$-7(5 - 1)^2 = 9 \neq 0 \pmod{11}.$$

Equations (7) and (8) therefore give a proper

$$SA^\top(22, 6, -, 5, 6 : 11 \times 12).$$

Theorem 7.1 (Main theorem). *For every odd prime power $q \equiv 3 \pmod{4}$ with $q \geq 11$, there exists a proper transposed sesqui array*

$$SA^\top\left(2q, \frac{q+1}{2}, -, \frac{q-1}{2}, \frac{q+1}{2} : q \times (q+1)\right).$$

Proof. The explicit parameters above cover $q = 11$. Every remaining admissible prime power is at least 19, so Proposition 6.4 provides switching parameters. Proposition 3.1, Theorem 4.1, and Proposition 5.2 then construct the required proper array. $\square$

Remark 7.2 (Lower parameters). The switching conditions themselves have no solution for $q = 3$ or 7: $\mathcal{A}$ is empty for $q = 3$; for $q = 7$, the sole candidate is $\alpha = 5$, and no r satisfies (2). This is a boundary of the present construction, not a nonexistence claim for all proper arrays with those parameters.

8 Prior-work boundary

The target parameter family overlaps substantially with earlier work. The Paley triple arrays of Preece, Wallis, and Yucas already have the same dimensions, replication, column concurrence, and row-column concurrence [7]. They have constant row concurrence; later work has studied their Youden-square representations and intercalates [6, 5]. Difference-set constructions give another infinite source of triple arrays, again with constant row concurrence by definition [4].

The deletion-exchange construction of Jäger et al. can transform a Youden rectangle into a row-column design and automatically preserves a binary, equireplicate column-balanced layer [3, Construction 5.2 and Theorem 5.3]. It does not automatically provide the row-column concurrence and properness proved here. Recent resolvable triple arrays also include an infinite Paley subfamily, but remain triple arrays [1].

Intercalates in Paley arrays are an established subject [5]. We therefore do not claim the Paley support, the target parameters, the character-sum method, or the Hasse bound as new. The contribution claimed here is the specific two-edge switch satisfying (1)–(2), the moment obstruction that makes its array proper, and the existence proof for every odd prime power $q \equiv 3 \pmod{4}$, $q \geq 11$.

The closest dated parameter anchor is the (22, 11, 12) entry in the 2025 table of Jäger et al. [2, Table 2]. That table is evidence of an open construction gap at the time of publication, not by itself proof of global priority. Accordingly, our novelty statement remains “to the best of our knowledge.”

The general Youden deletion interface does not imply this construction. Whether individual arrays from the present family also admit alternative compatible Youden representations is a separate representation problem and is not used in the proof.

9 Reproducibility

The proof is independent of computation. The accompanying repository contains five complementary verification paths:

- a symbolic check of the first- and second-moment identities;
- an exact check of the character-count formula over sample prime fields;
- full cellwise array checks over sample prime fields;
- an implementation-independent check over $\text{GF}(27)$; and
- a generic finite-field audit covering positive and negative boundary cases.

The corresponding scripts are


```

verify_general_paley_switch_symbolic.py
verify_general_paley_switch_character_count.py
verify_general_paley_switch_prime_fields.py
verify_general_paley_switch_gf27_independent.py
audit_general_paley_switch_generic_fields.py

```

They test the formulas and finite instances; they are not used to extrapolate the infinite conclusion in Theorem 7.1.

Declaration of AI-assisted research and writing

This research program used OpenAI Codex extensively for candidate selection, conjecture formation, finite-field derivations, adversarial proof checking, literature-search assistance, verification-code generation, and manuscript drafting and revision. The retained claims are presented through human-readable arguments and reproducible artifacts, but AI output is not treated as an authority and independent expert review remains pending. Codex is a tool, not an author. The named author, Carptopus, accepts responsibility for the manuscript’s accuracy, originality, citations, and integrity.

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